

SQL Project:- Exploring Customer Behavior

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Batch:-164 Business Analysis

Introduction

In today's competitive electronics retail industry, understanding customer purchasing behavior is essential for improving sales performance and customer satisfaction. This project focuses on analyzing customer transaction data from an electronics store using SQL. The database consists of customer information, product details, sales transactions, and premium membership data.

By analyzing these datasets, we can identify purchasing patterns, high-value customers, top-selling products, revenue trends, and the effectiveness of the Gold Membership Program. The insights generated from this analysis can help businesses make data-driven decisions regarding marketing strategies, inventory management, customer retention, and loyalty programs.

Objective of the Project

The main objectives of this project are:

- To analyze customer purchasing behavior using SQL queries.
- To identify the highest revenue-generating products.
- To determine the most valuable customers based on spending patterns.
- To evaluate the performance of the Gold Membership Program.
- To analyze yearly sales and revenue trends.
- To identify popular products among customers.
- To measure customer engagement through purchase frequency.
- To generate business insights that support decision-making and improve customer retention.

Dataset Description:-

This project uses five datasets:

1. Users

Contains customer registration information including User ID and Signup Date.

2. User_Name

Contains User ID and corresponding customer names.

3. Products

Contains electronic product details such as Product ID, Product Name, and Price.

4. Sales

Contains transaction records including User ID, Product ID, and Purchase Date.

5. GoldUsers_Signup

Contains information about customers who subscribed to the Gold Membership Program and their membership start date.

Tools and Technologies Used

- SQL Server Management Studio (SSMS)

- SQL Queries
- Joins
- Aggregate Functions
- Window Functions
- Date Functions
- Data Analysis Techniques

Q-1 Total Sales Revenue Generated by Each Product?

The screenshot shows the Microsoft SQL Server Enterprise Manager interface. The Object Explorer on the left shows the server structure. The central pane displays a SQL query in the 'SQLQuery2.s...CV\Hp (80)' window. The query calculates the total revenue for each product by joining the Products and Sales tables and grouping by product name. The results are shown in a table at the bottom.

```

(9, '2018-12-01');

SELECT * FROM Products;
SELECT * FROM Sales;
SELECT * FROM Users;
SELECT * FROM User_Name;
SELECT * FROM GoldUsers_Signup;

SELECT
    p.product_name,
    SUM(p.price) AS Total_Revenue
FROM Sales s
INNER JOIN Products p
ON s.product_id = p.product_id
GROUP BY p.product_name
ORDER BY Total_Revenue DESC;

```

	product_name	Total_Revenue
1	iPhone 15	640000
2	Samsung Galaxy S24	300000
3	Dell Laptop	260000
4	Apple Watch	175000
5	HP Laptop	120000
6	Sony Headphones	90000
7	Tablet	60000
8	Gaming Keyboard	16000
9	Bluetooth Speaker	15000
10	Wireless Mouse	3000

Query executed successfully. DESKTOP-017PMC\SQLEXPRESS ... DESKTOP-017PMC\Hp (80) customerbehaviour1 | 00:00:00 | Row: 1, Col: 1 | 10 rows

Key Findings:-

- The highest revenue was generated by Iphone 15
- Premium electronic products contributed significantly to total sales revenue.
- Products such as wireless mouse generated comparatively lower revenue.
- Revenue distribution indicates that customers prefer high-value electronic products.
- The company should focus marketing efforts on top-performing products while improving sales strategies for lower-performing products.

Q2. Which 3 Products Have the Highest Sales Revenue?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DESKTOP-017PMC\SQLEXPRESS (SQL Server 17.0.1115)'. The main window shows a SQL query in the 'SQLQuery2.s...CV\Hp (801)*' tab. The query calculates the total revenue for the top 3 products. The results pane at the bottom shows the following data:

product_name	Total_Revenue
1 iPhone 15	640000
2 Samsung Galaxy S24	300000
3 Dell Laptop	260000

Key Findings:-

- The top 3 products i.e., (iPhone 15, Samsung Galaxy S24 and Dell Laptop) contributed the highest share of total revenue.
- These products represent the most profitable product categories in the business.
- The company should ensure sufficient inventory availability for these products.
- Marketing campaigns should prioritize these products to maximize revenue growth.
- These products can be used as flagship offerings to attract more customers.

Q3. How Many Users Have Signed Up for the Service and Taken Gold Membership?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DESKTOP-017PMC\SQLEXPRESS (SQL Server 17.0.1115)'. The main window shows a SQL query in the 'SQLQuery1.s...CV\Hp (52)*' tab. The query counts the number of users and gold members. The results pane at the bottom shows the following data:

(No column name)
1 10

(No column name)
1 5

Key Findings:-

- A total of 10 customers registered on the platform.
- Out of these, 5 customers upgraded to Gold Membership.

- The Gold Membership adoption rate is 50%.
- This indicates a good level of customer interest in the premium membership program.
- The business can further improve membership adoption through exclusive offers and loyalty rewards.

Approximately 50% of registered users subscribed to the Gold Membership Program, indicating moderate customer engagement with premium services.

Q4. What is the Revenue Generated from Gold Users?

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with a tree view of the database structure, including 'Databases', 'System Databases', 'Database Snapshots', 'customer_behavior', 'CustomerBehavior', 'customerbehaviour1', 'Database Diagrams', 'Tables', 'Views', 'External Resources', 'Synonyms', 'Programmability', 'Query Store', 'Service Broker', 'Storage', 'Security', 'HR_DB', 'interviewpractise', 'SQLBatch_21stMar', 'WorkerAnalysis', 'WorkforceAnalysis', 'Security', 'Server Objects', 'Replication', 'Management', and 'XEEvent Profiler'. The right pane shows a SQL query window with the following code:

```

1  SELECT * FROM Products;
2  SELECT * FROM Sales;
3  SELECT * FROM Users;
4  SELECT * FROM User_Name;
5  SELECT * FROM GoldUsers_Signup;
6
7  (select count(*) from Users)
8  (select count(*) from GoldUsers_Signup)
9
10 -----revenue from goldusers-----
11 SELECT
12     SUM(p.price) AS Revenue_From_Gold_Users
13 FROM Sales s
14 INNER JOIN GoldUsers_Signup g
15     ON s.user_id = g.user_id
16 INNER JOIN Products p
17     ON s.product_id = p.product_id;

```

The query results are displayed in a table with the following data:

Revenue_From_Gold_Users
975000

Key Findings:-

- Gold Members contributed a significant portion of total sales revenue.
- Premium members demonstrated higher purchasing activity compared to regular customers.
- The Gold Membership Program appears to positively influence customer spending behavior.
- Revenue generated by Gold Members highlights the importance of customer loyalty programs.
- The company should continue investing in Gold Membership benefits to increase customer retention and revenue.

Q5. What is the Frequency of Customer Purchases?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'customerbehaviour1'. The main window shows a SQL query titled 'SQLQuery1.s...CV\Hp (52)' with the following code:

```

10  -----revenue from goldusers-----
11  SELECT
12  SUM(p.price) AS Revenue_From_Gold_Users
13  FROM Sales s
14  INNER JOIN GoldUsers_Signup g
15  ON s.user_id = g.user_id
16  INNER JOIN Products p
17  ON s.product_id = p.product_id;
18
19  SELECT
20  u.user_name,
21  COUNT(*) AS Total_Orders
22  FROM Sales s
23  INNER JOIN User_Name u
24  ON s.user_id = u.user_id
25  GROUP BY u.user_name
26  ORDER BY Total_Orders DESC;

```

The Results pane shows the output of the second query, listing the top 10 users by total orders:

user_name	Total_Orders	
1	Amit	4
2	Anjali	4
3	Karan	4
4	Neha	4
5	Pooja	4
6	Priya	4
7	Rahul	4
8	Rohit	4
9	Sneha	4
10	Vikas	4

Key Findings:-

- The analysis identified the most active customers on the platform.
- Frequent purchasers demonstrate strong customer engagement.
- Some customers purchase significantly more often than others.
- Highly active customers may be suitable candidates for loyalty rewards.
- Purchase frequency is an important indicator of customer retention.

Q. What Is the Most Purchased Product on the Platform?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'customerbehaviour1'. The main window shows a SQL query titled 'SQLQuery1.s...CV\Hp (52)' with the following code:

```

20  u.user_name,
21  COUNT(*) AS Total_Orders
22  FROM Sales s
23  INNER JOIN User_Name u
24  ON s.user_id = u.user_id
25  GROUP BY u.user_name
26  ORDER BY Total_Orders DESC;
27
28
29  SELECT TOP 1
30  p.product_name,
31  COUNT(*) AS Purchase_Count
32  FROM Sales s
33  INNER JOIN Products p
34  ON s.product_id = p.product_id
35  GROUP BY p.product_name
36  ORDER BY Purchase_Count DESC;

```

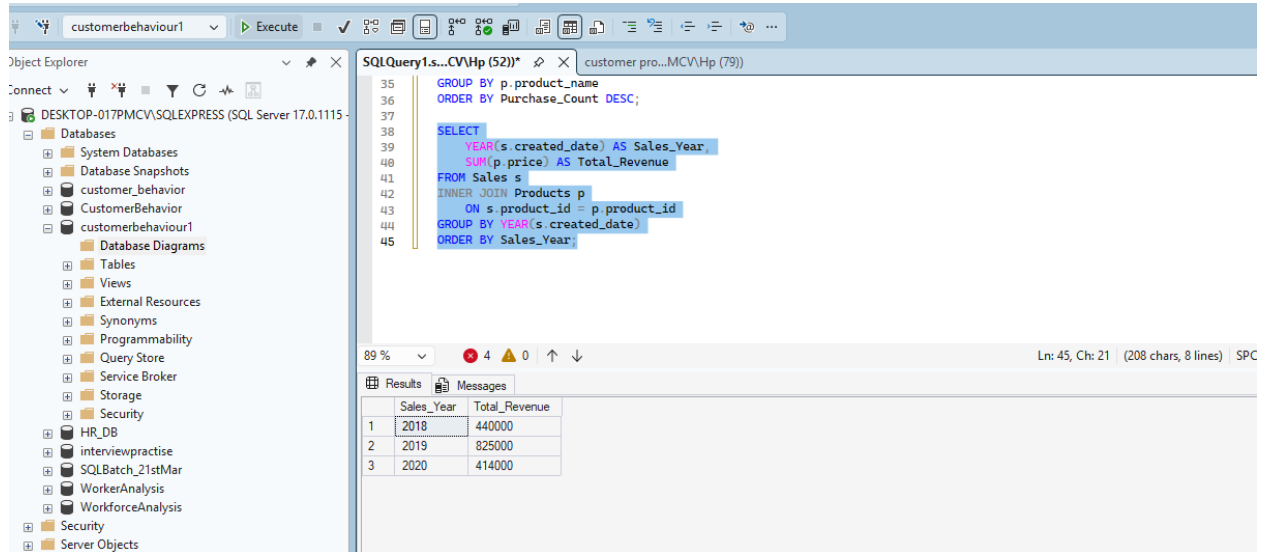
The Results pane shows the output of the second query, listing the top 1 product by purchase count:

product_name	Purchase_Count	
1	iPhone 15	8

Key Findings

- Customers show strong preference toward this product.
- The product contributes significantly to overall sales volume.
- The company should ensure adequate stock availability.
- Marketing campaigns can leverage this product to attract customers.

Q. What Is the Total Sales Revenue Generated in Each Year?



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure, including the 'customerbehaviour1' database. The central pane shows a SQL query being executed. The query is as follows:

```

35 GROUP BY p.product_name
36 ORDER BY Purchase_Count DESC;
37
38 SELECT
39     YEAR(s.created_date) AS Sales_Year,
40     SUM(p.price) AS Total_Revenue
41 FROM Sales s
42 INNER JOIN Products p
43     ON s.product_id = p.product_id
44 GROUP BY YEAR(s.created_date)
45 ORDER BY Sales_Year;

```

The Results pane at the bottom displays the output of the query, showing the total revenue for each year from 2018 to 2020.

Sales_Year	Total_Revenue
2018	440000
2019	825000
2020	414000

Key Findings;-

- The analysis reveals annual sales performance.
- Revenue trends help identify business growth patterns.
- High-performing years indicate strong customer demand.
- Revenue fluctuations may highlight market opportunities or challenges.
- Yearly trend analysis supports future planning and forecasting.

Q. How Many Products Were Sold?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure, including the 'customerbehaviour1' database. The SQL Query window on the right contains the following query:

```
GROUP BY p.product_name
ORDER BY Purchase_Count DESC;

SELECT
YEAR(s.created_date) AS Sales_Year,
SUM(p.price) AS Total_Revenue
FROM Sales s
INNER JOIN Products p
ON s.product_id = p.product_id
GROUP BY YEAR(s.created_date)
ORDER BY Sales_Year;

select count(*) as total_product_sold
from Sales
```

The Results pane at the bottom shows the output of the query:

total_product_sold
40

Key Findings:-

- Total products sold on the platform were 40.
- The company has processed a significant number of transactions.
- Sales volume indicates customer activity on the platform.

Q. What Is the Average Price of Products?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure, including the 'customerbehaviour1' database. The SQL Query window on the right contains the following query:

```
SELECT
YEAR(s.created_date) AS Sales_Year,
SUM(p.price) AS Total_Revenue
FROM Sales s
INNER JOIN Products p
ON s.product_id = p.product_id
GROUP BY YEAR(s.created_date)
ORDER BY Sales_Year;

select count(*) as total_product_sold
from Sales

select AVG(price) as Avg_product_price
from Products
```

The Results pane at the bottom shows the output of the query:

Avg_product_price
33863

Key Findings

- The average product price is ₹33863.
- This helps understand the overall pricing strategy.
- The company offers products across different price ranges.

Q. Which Is the Most Expensive Product?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DESKTOP-017PMC\SQLEXPRESS (SQL Server 17.0.1115)'. The central pane shows a SQL query window with the following code:

```

ORDER BY Sales_Year;

select count(*) as total_product_sold
from Sales

Select AVG(price) as Avg_product_price
from Products

select top 1 product_name,price
from Products
order by price desc;

```

The query results are displayed in a table below the code:

product_name	price
1 iPhone 15	80000

Key Findings:-

- iPhone 15 is the most expensive product.
- Premium products contribute significantly to revenue.
- High-value products attract customers seeking premium electronics.

Q. Which Is the Cheapest Product?

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DESKTOP-017PMC\SQLEXPRESS (SQL Server 17.0.1115)'. The central pane shows a SQL query window with the following code:

```

from Sales

Select AVG(price) as Avg_product_price
from Products

select top 1 product_name,price
from Products
order by price desc;

select top 1 product_name,price
from Products
order by price Asc;

```

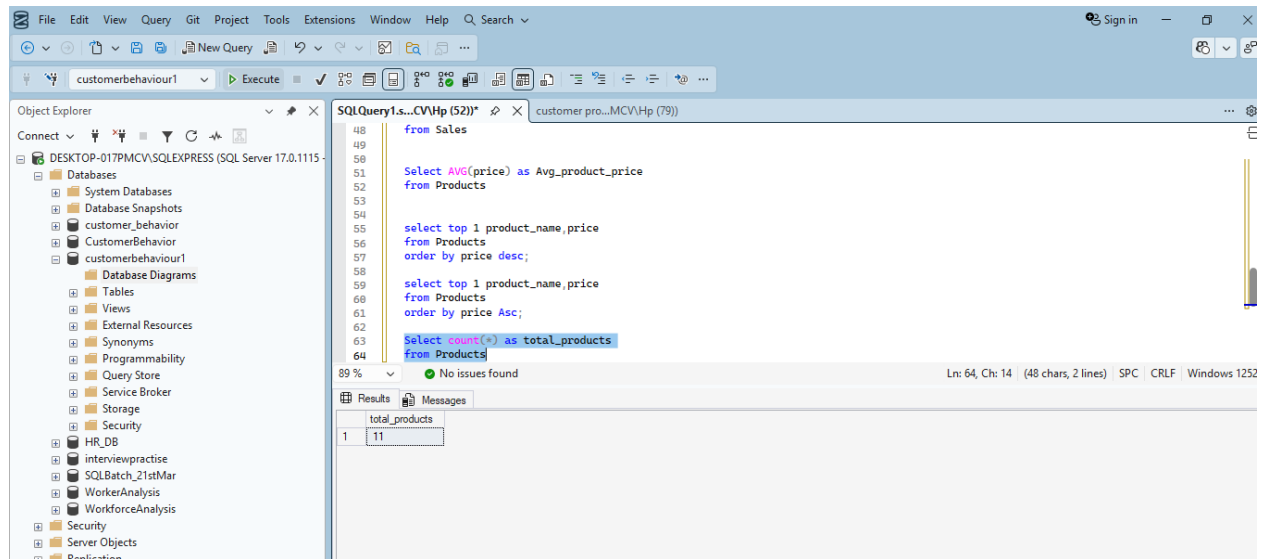
The query results are displayed in a table below the code:

product_name	price
1 Wireless Mouse	1500

Key Findings

- Wireless is the least expensive product.
- Affordable products help attract budget-conscious customers.
- Low-cost products can increase transaction volume.

Q-How Many Products Are Available?



Key Findings

- The store offers 11 products.
- Product variety helps serve different customer needs.
- A diverse catalog improves customer choice.

Conclusion: -

This project successfully analyzed customer purchasing behavior using SQL and provided valuable insights into sales performance, customer engagement, and product popularity. By examining customer transactions, product information, and membership data, meaningful business intelligence was generated.

The analysis identified top-selling products, high-spending customers, purchase frequency patterns, and revenue trends. These insights can help businesses improve decision-making, optimize inventory management, strengthen customer relationships, and increase profitability.

The project also demonstrated the practical application of SQL concepts such as Joins, Aggregate Functions, Group By, Sorting, and Filtering in solving real-world business problems. Overall, this analysis highlights the importance of data-driven decision-making in understanding customer behavior and improving business performance.

Thankyou